Towards comparing different radiation methods used in astrophysical codes

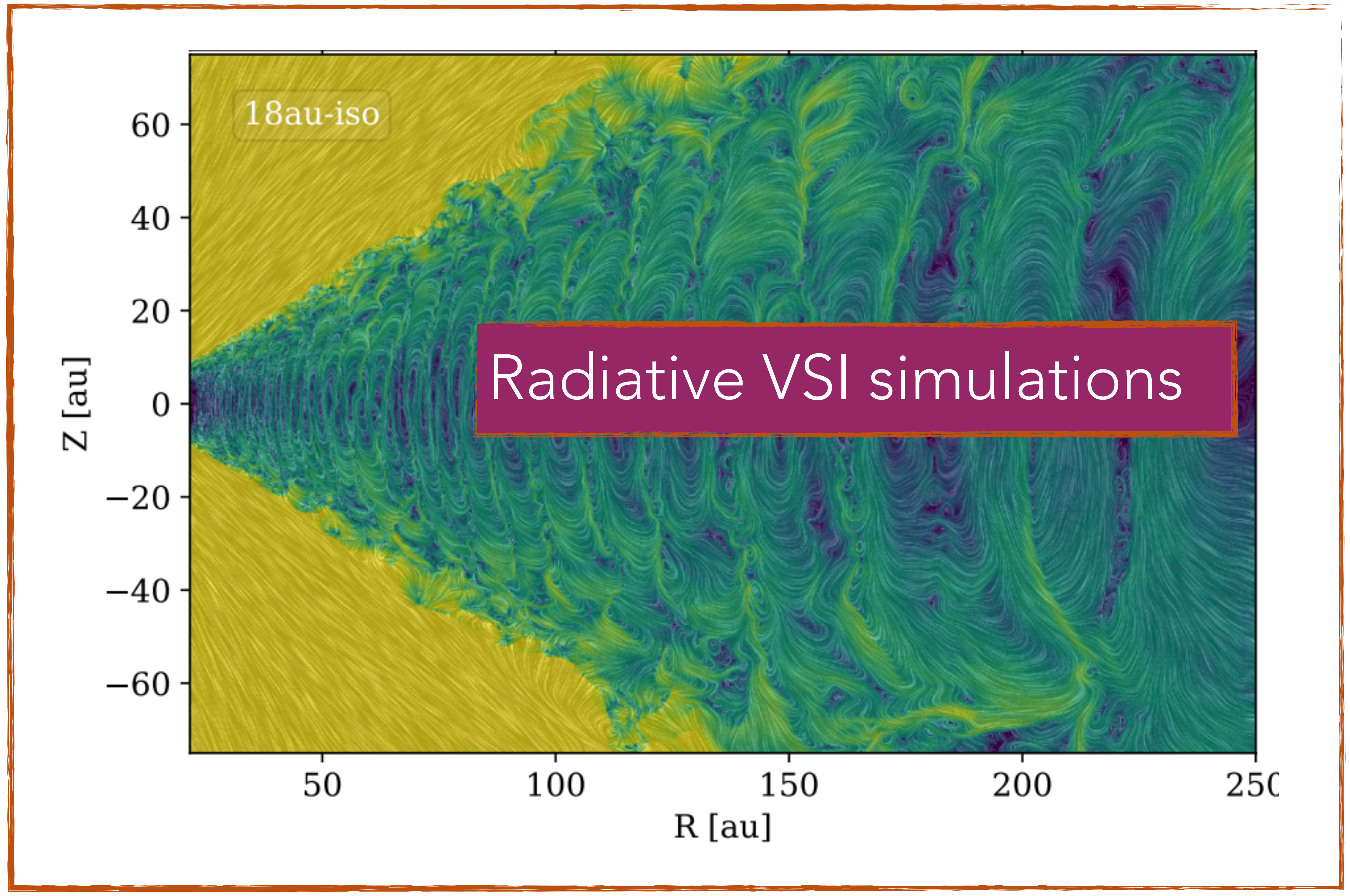
in the context of protoplanetary disks

Kruti Sudarshan (PhD student, Max Planck Institute for Astronomy)
sudarshan@mpia.de, prakrutisudarshan.github.io

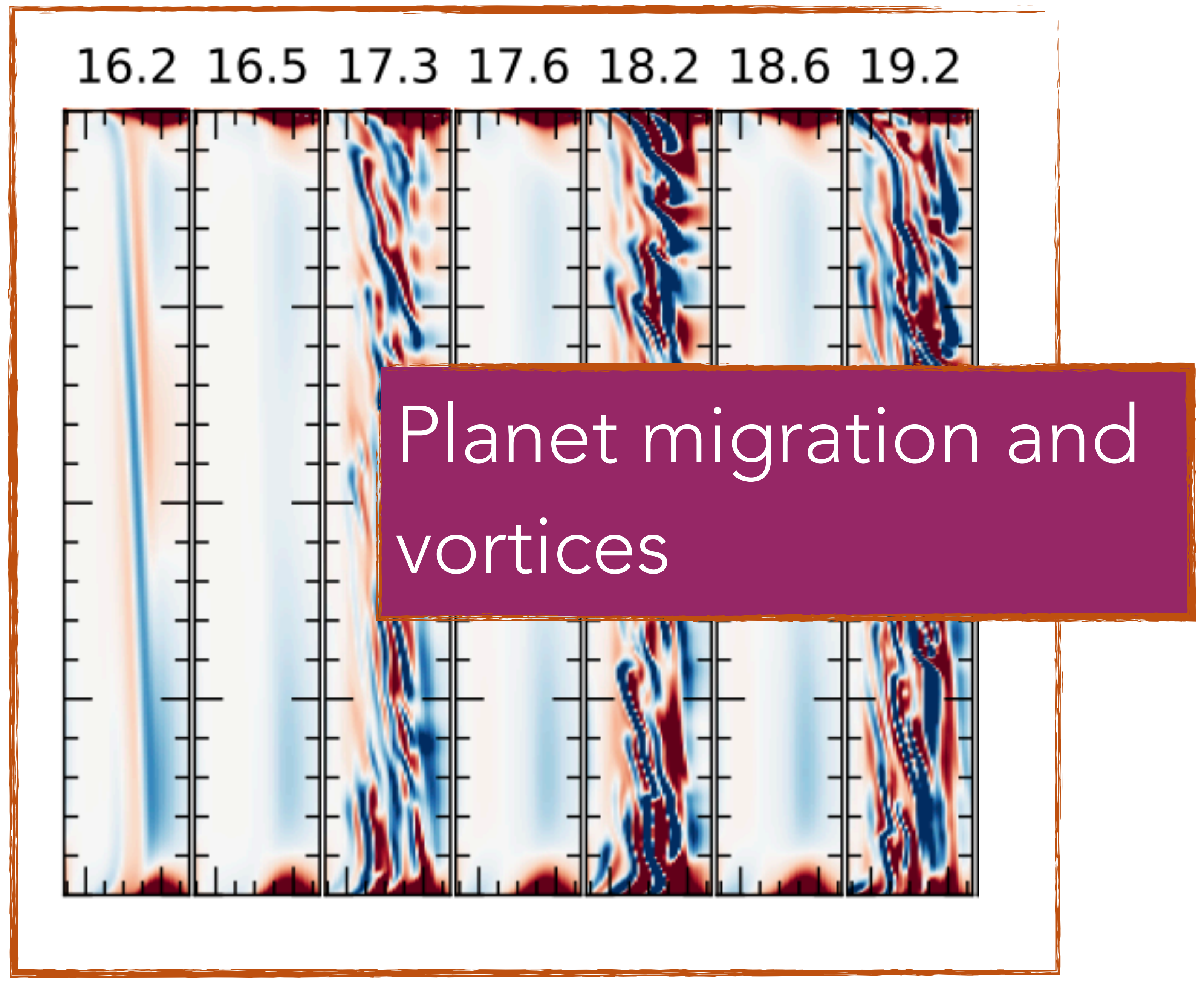
M1, half-moment simulations in this presentation are by David Melon Fuksman
thanks to Alex Ziampras, Mario Flock, Stanley Baronett, Thomas Pfeil, Zhaohuan Zhu

PFITS+ meeting, June 12 2026

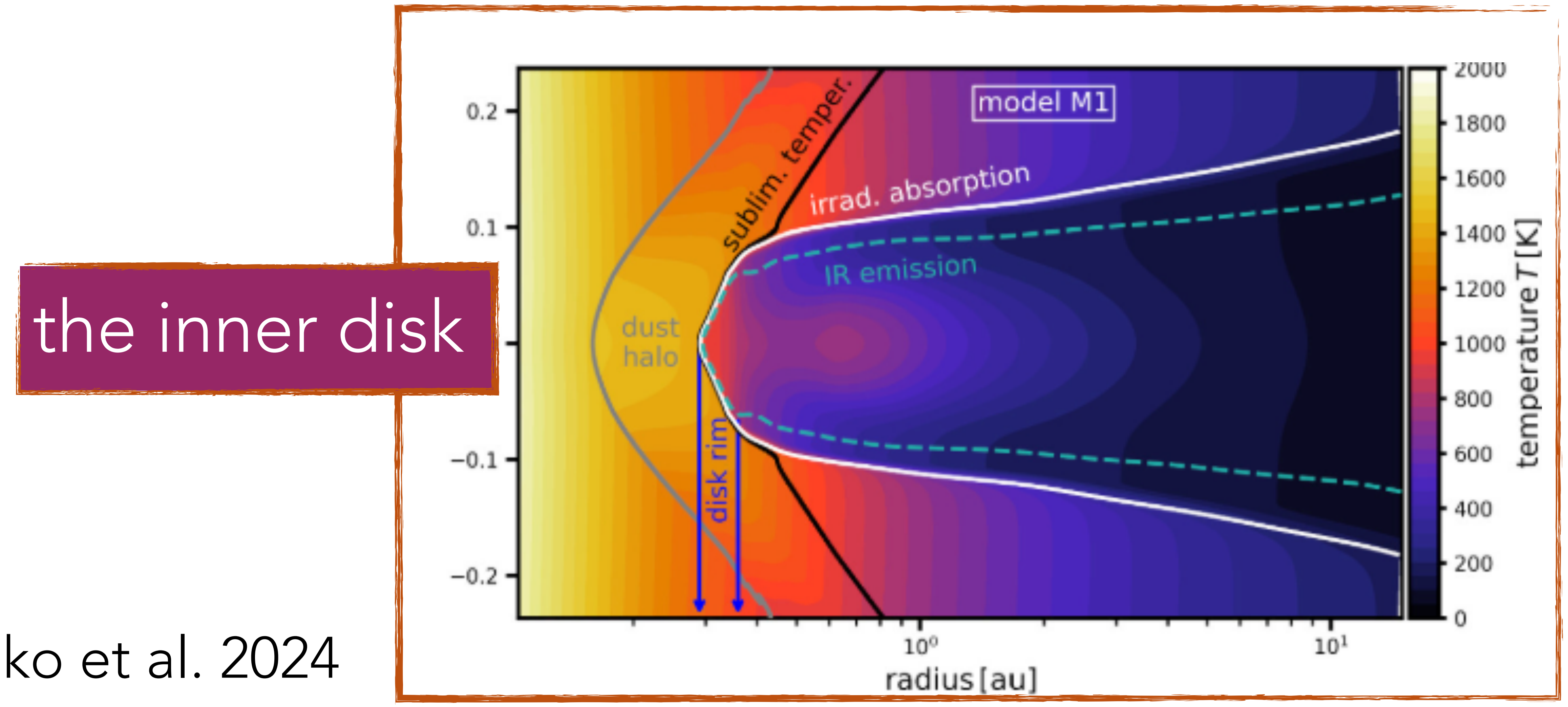
Radiation controls a lot of processes in protoplanetary disks.



Zhang et al. 2024



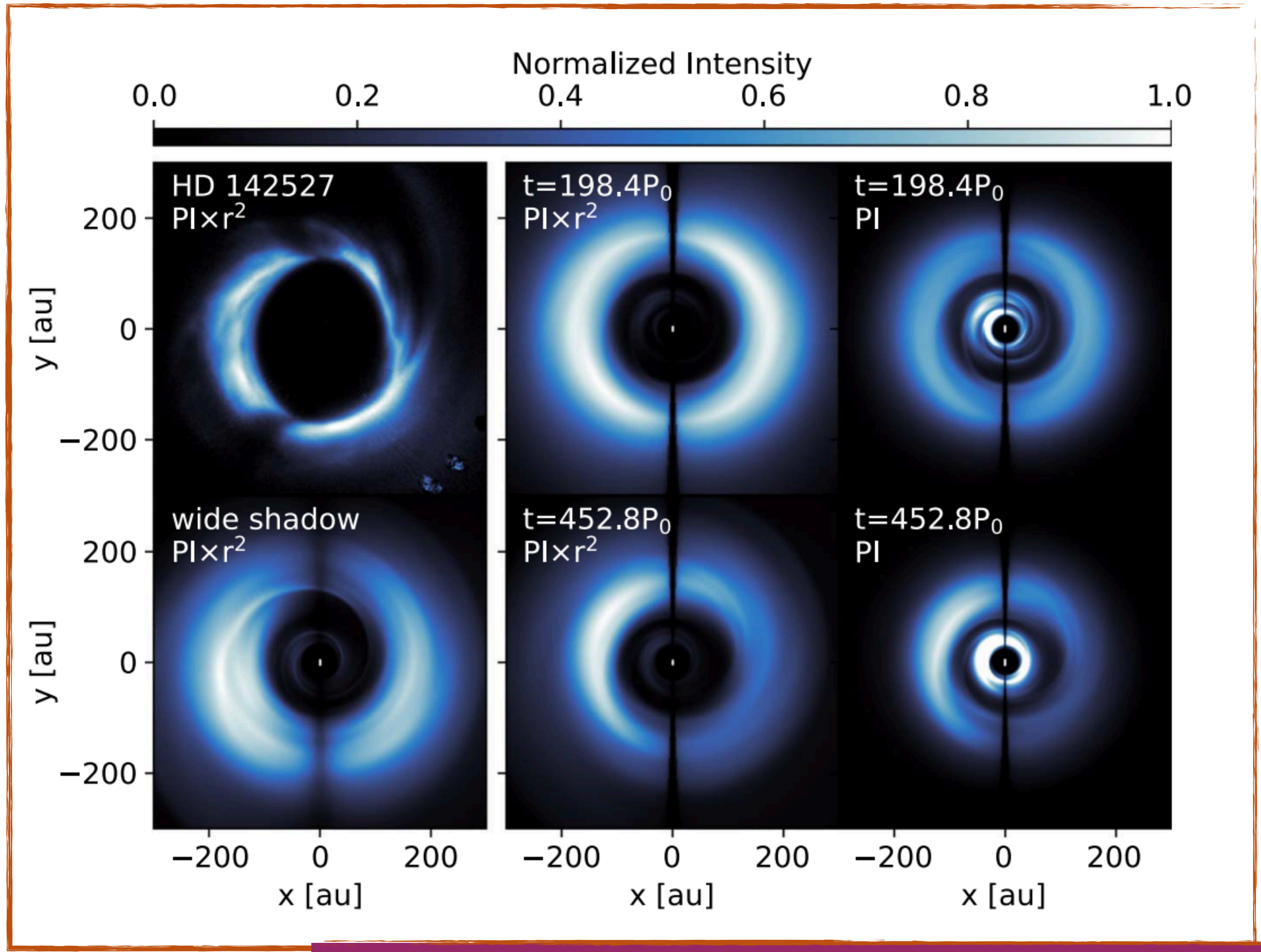
Ziampras et al. 2025

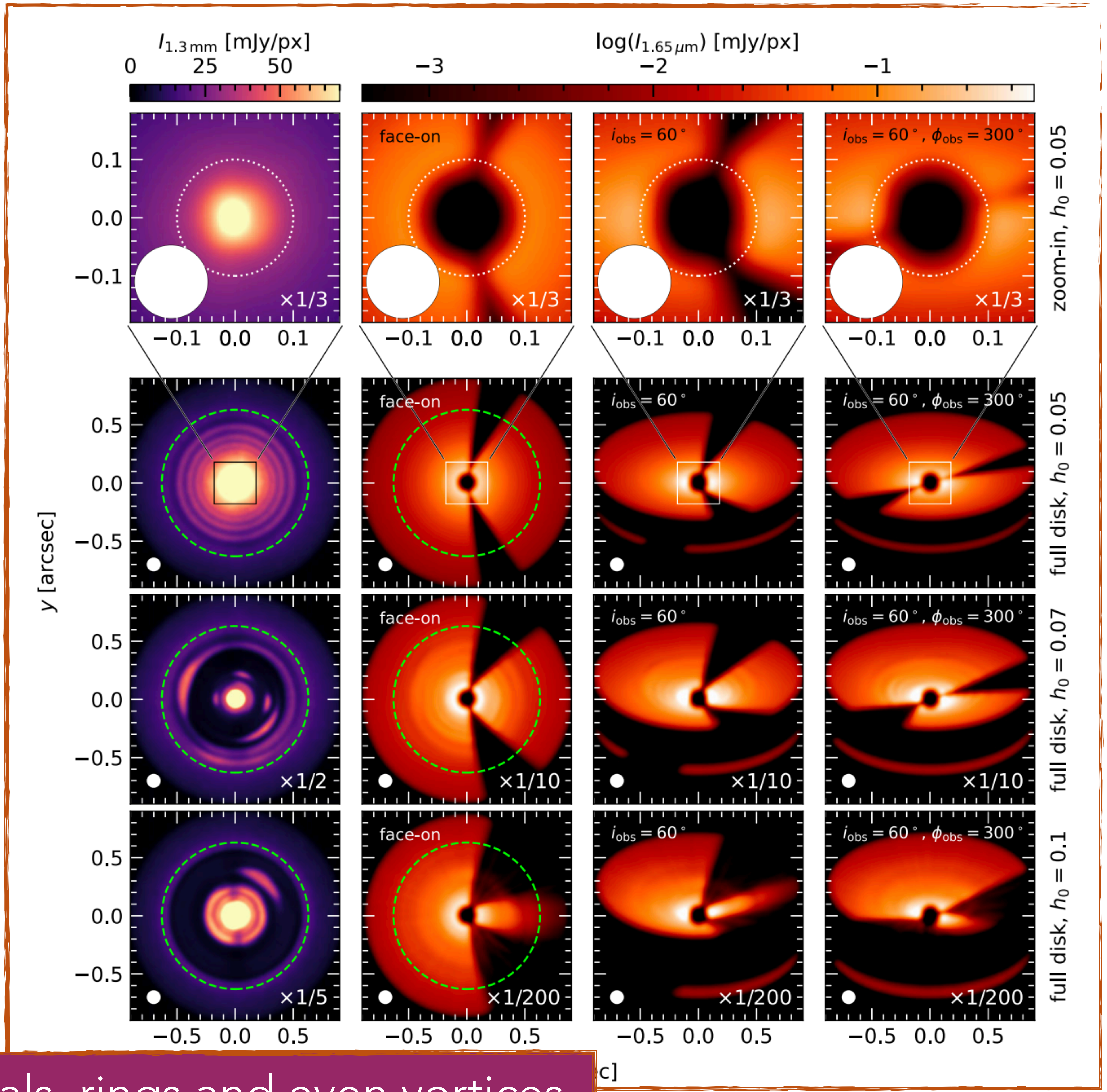


Chrenko et al. 2024

Thermally driven substructures:

Su & Bai 2024, Zhang+24, 25, Muley+25, Ziampras+25

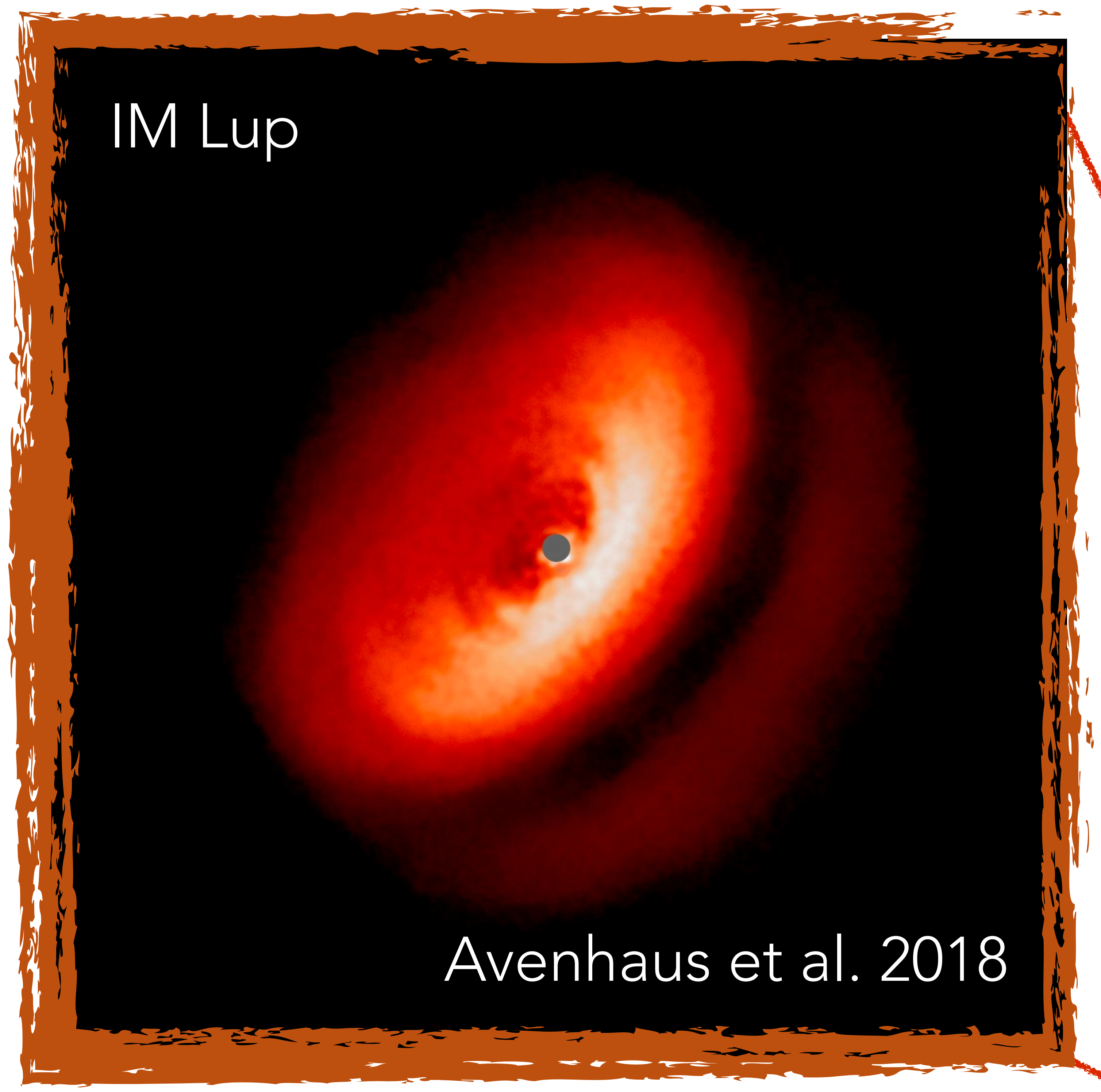


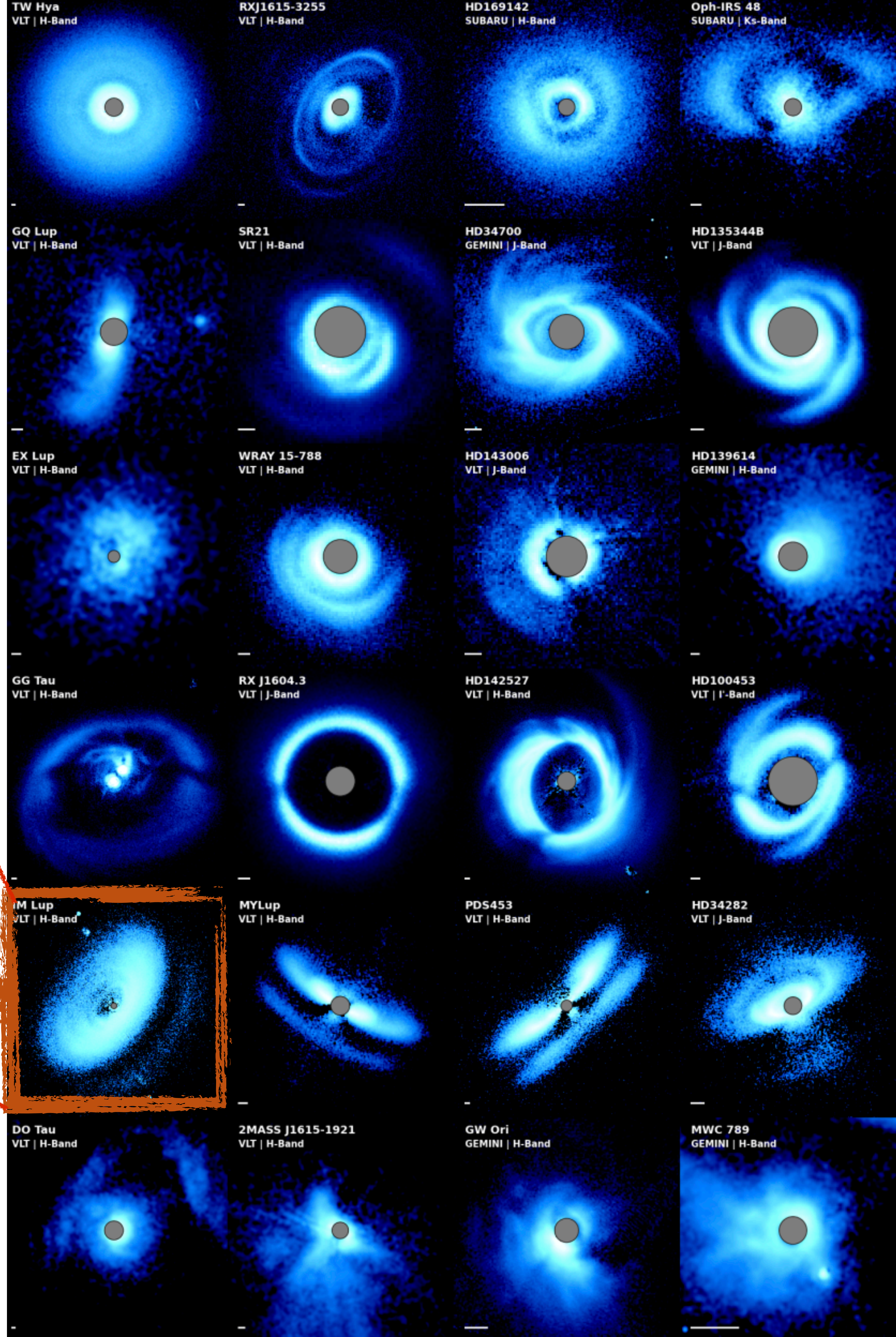


Misaligned disks can thermally drive spirals, rings and even vortices

My PhD work: radiation hydrodynamics of starlight-driven self-shadowing

Flared disk profile





Benisty et al. 2023 PPVII Chapter

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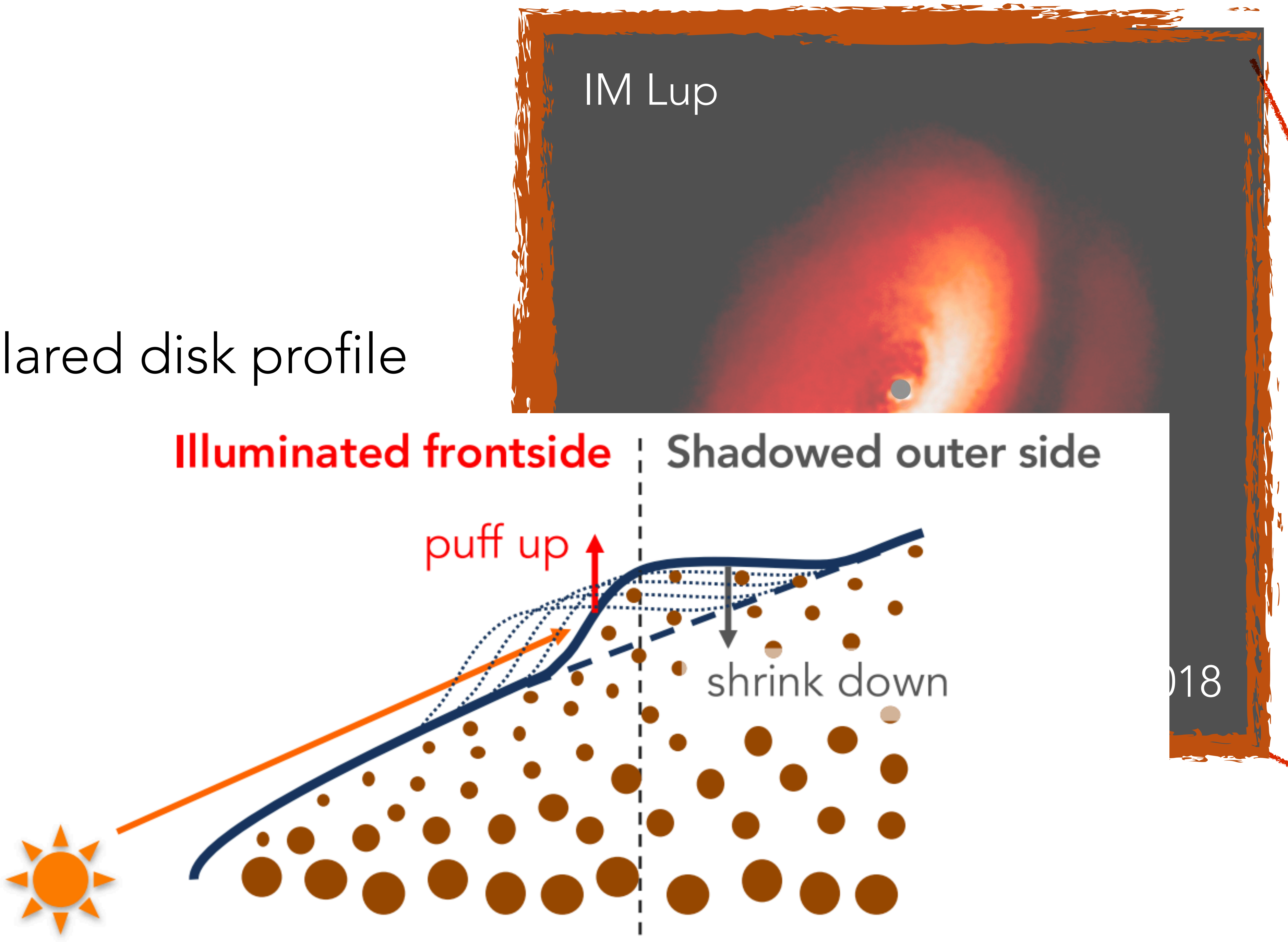
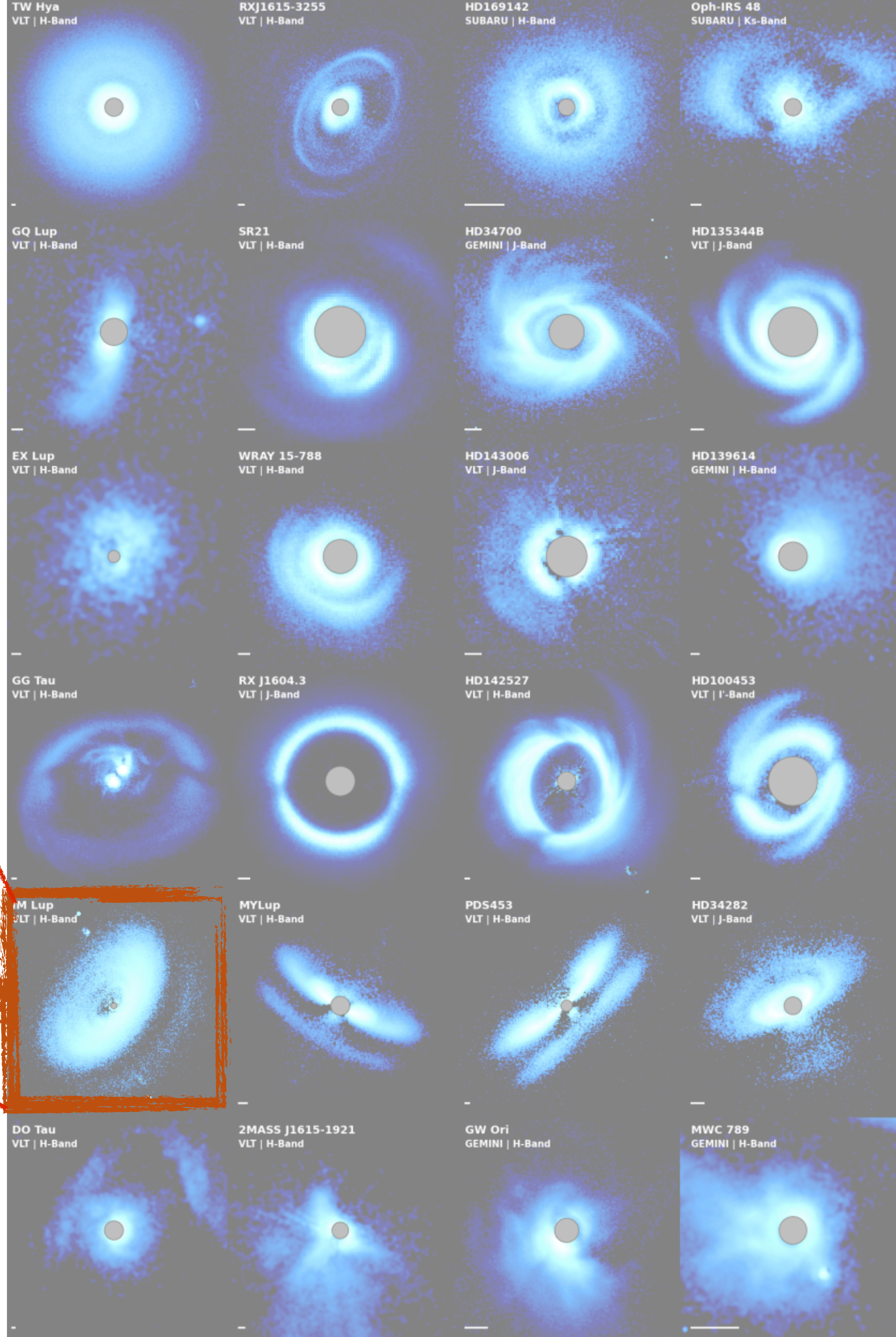
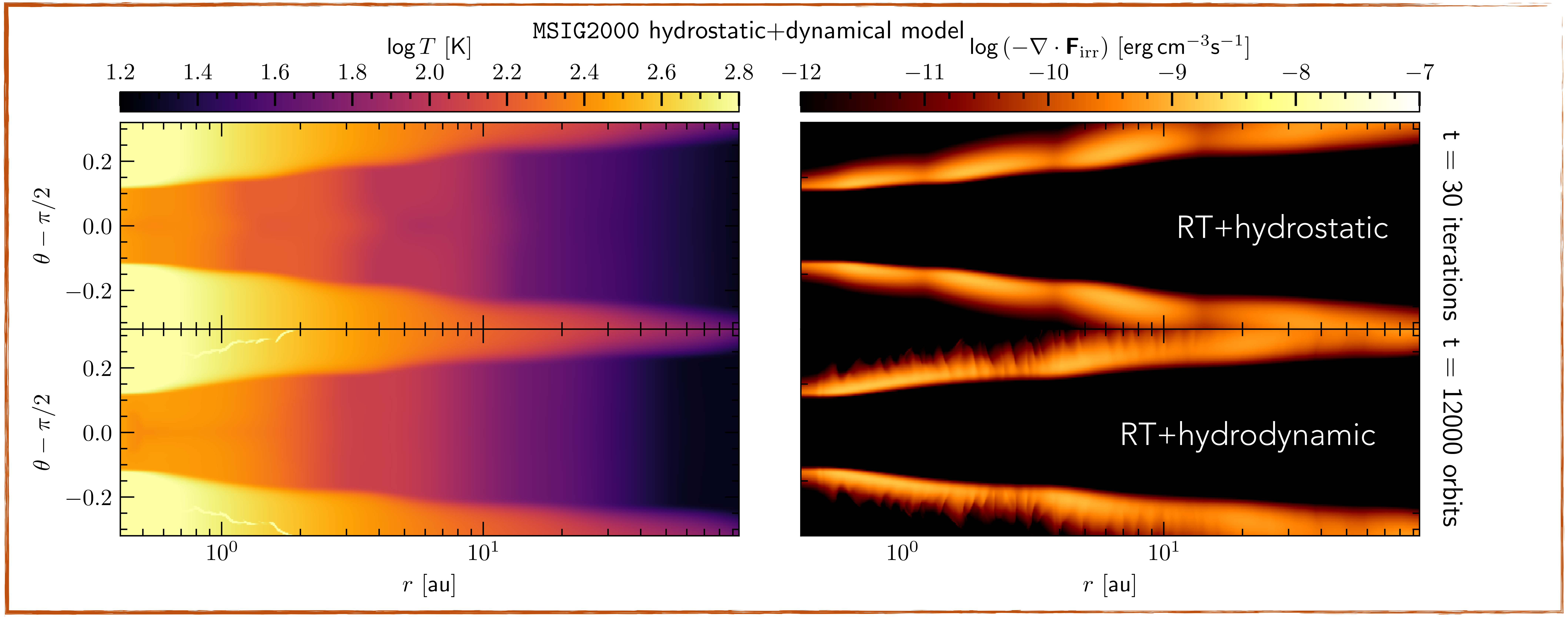




Ueda et al. 2021 (also Dullemond 2000, Watanabe & Lin 2008)

My PhD work: radiation hydrostatics and dynamics of self-shadowed disks

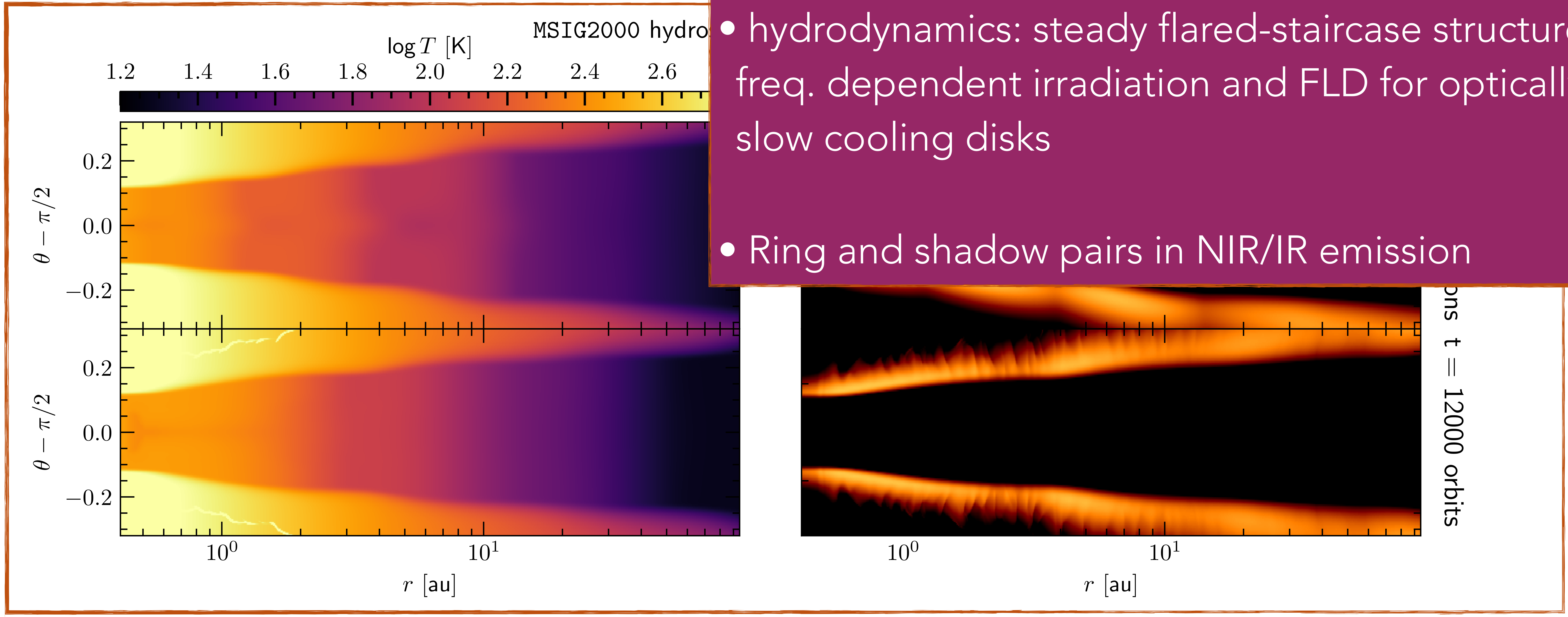
Optically thick disks, d2g = 10^{-2}



Sudarshan et al. 2026

My PhD work: radiation hydrostatics and dynamics of self-shadowed disks

- hydrostatics: thermal wave instability, inwardly moving thermal waves
- hydrodynamics: steady flared-staircase structure with freq. dependent irradiation and FLD for optically thick, slow cooling disks
- Ring and shadow pairs in NIR/IR emission



Sudarshan et al. 2026

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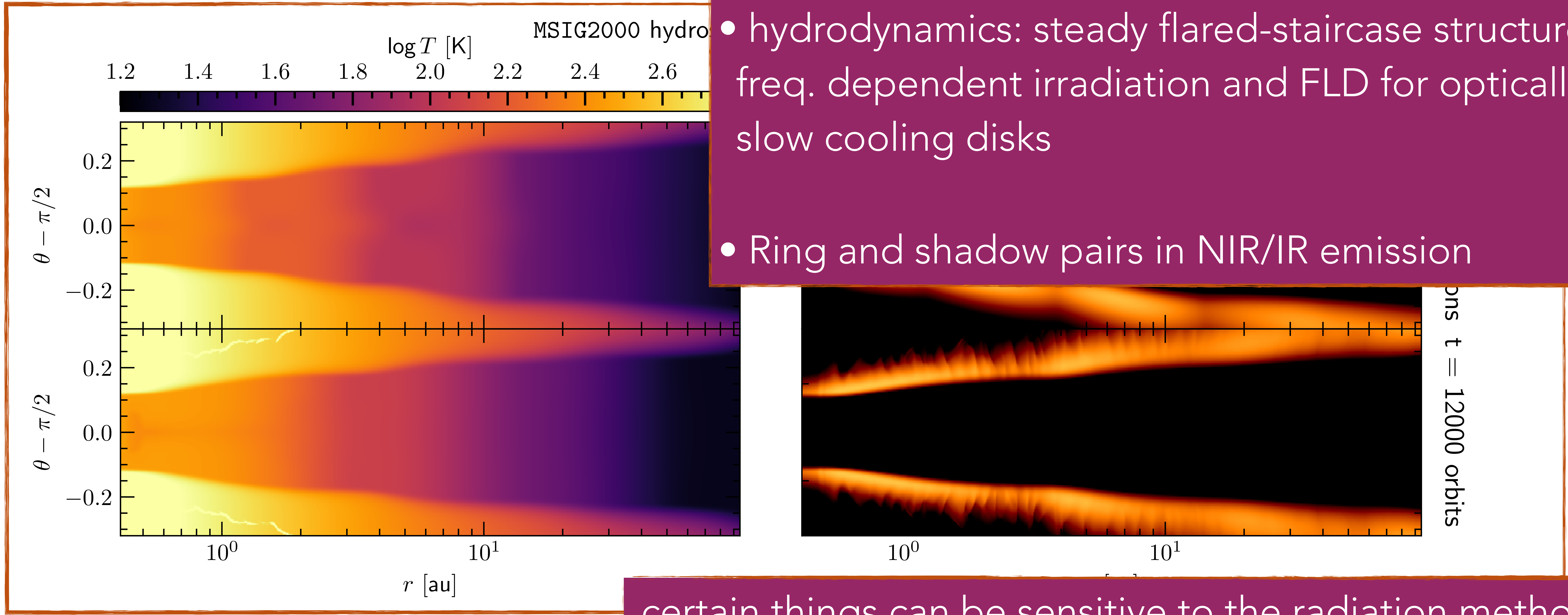


- hydrostatics: thermal wave instability, inwardly moving thermal waves

- hydrodynamics: steady flared-staircase structure with freq. dependent irradiation and FLD for optically thick, slow cooling disks

- Ring and shadow pairs in NIR/IR emission

certain things can be sensitive to the radiation method you use!

Sudarshan et al. 2026

numerical methods are improving! (these are ones from recent years)

Frequency-dependent discrete ordinates (Baronett et al. 2026)

Half-moment closure (Melon Fuksman et al. 2025)

Multi-group FLD (Robinson et al. 2024)

Three-temperature (Muley et al. 2023)

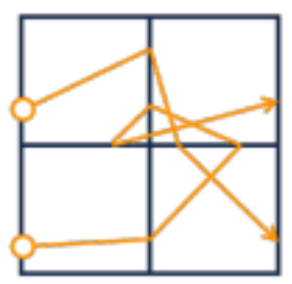
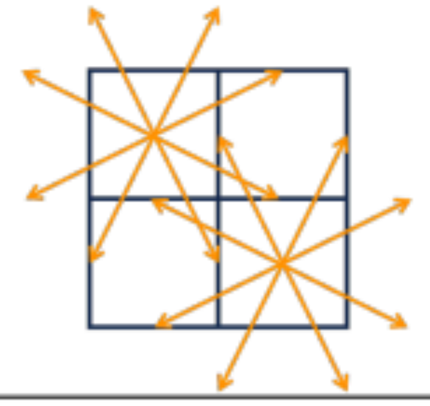
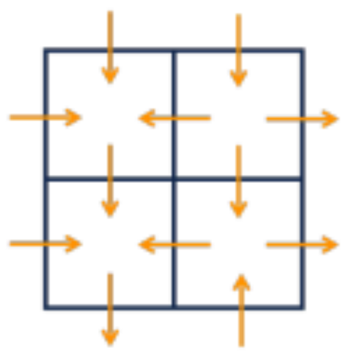
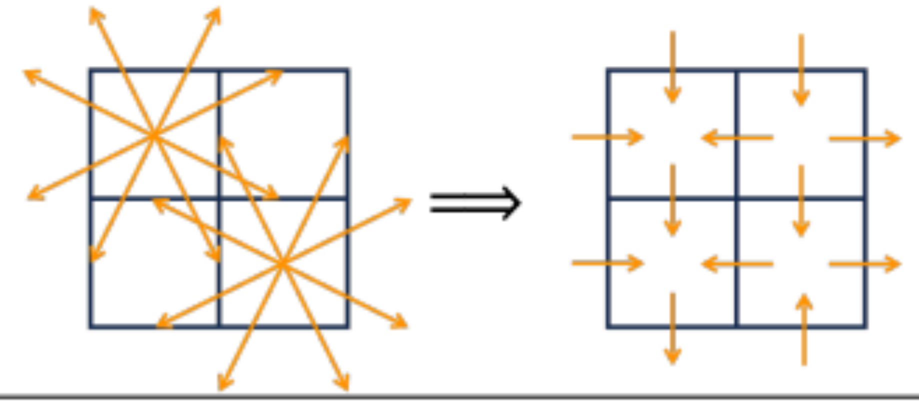
More physics ↑

More cost ↑↑↑↑

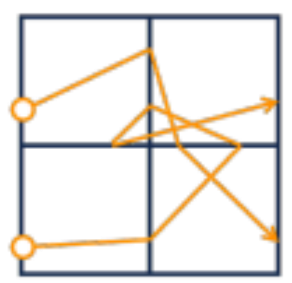
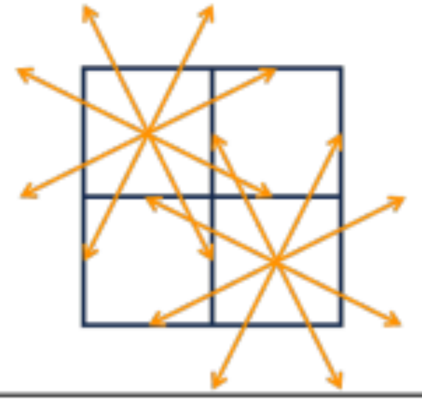
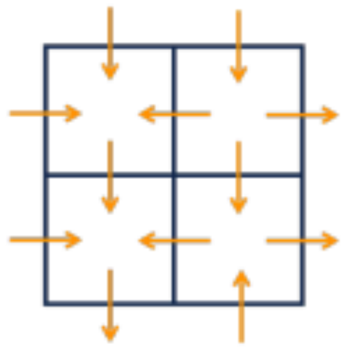
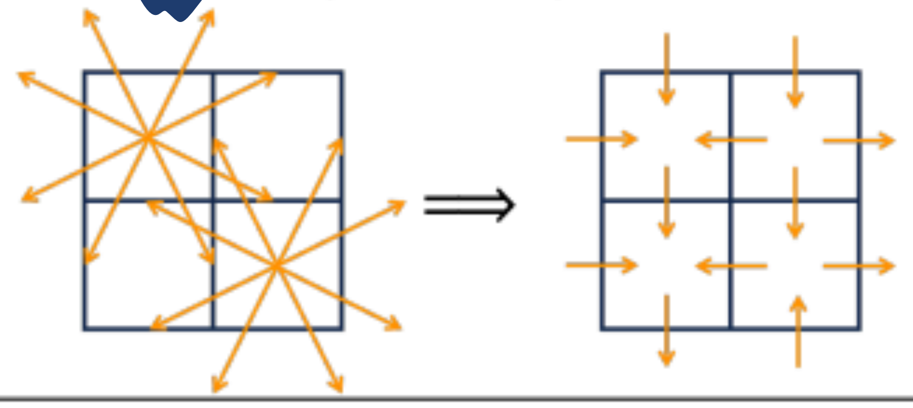
Coupling with dust dynamics, coagulation, and additional physics still requires you to work with simpler moments.

We are using different radiation methods.

Q: what are the differences, in realistic disk setups?

Category	Particle	Ray	Fluid	Ray-fluid hybrid
				
Method name	Monte Carlo	Ray-tracing Characteristics	Flux-limited diffusion M1 closure	Variable Eddington tensor Discrete ordinates
Physics				
Accurate direction of radiation?	✓	✓	× ($\tau \lesssim 1$)	✓
Accurate at different optical depths τ?	× ($\tau \gg 1$)	✓	× ($\tau \lesssim 1$)	✓
Time-and-velocity-dependent?	✓	×	✓	✓
Detailed microphysics (e.g. anisotropic scattering)?	✓	×	×	×
Known difficulties	Poisson noise; Slow transport ($\tau \gg 1$)	Ray-effect* ($\tau \ll 1$)	Filling shadows; Merging crossing beams; Colliding radiation fronts	Ray-effect* ($\tau \ll 1$)
Computational cost for a sophisticated grey 3D simulation (1= comparable to hydro, order-of-magnitude estimate)				
Relative cost	×100	×10	×1	×10

Ma et al. 2025

Category	Particle	Ray	Fluid	Ray-fluid hybrid
				
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Ma et al. 2025

Moment methods: FLD and M1

Radiation intensity $$\frac{1}{c}\frac{\partial I}{\partial t} + \mathbf{n} \cdot \nabla I = \eta - \chi I$$

Moments:

$$E_r = \frac{1}{c}\int I \, d\Omega \qquad \mathbf{F}_r = \int I \, \mathbf{n} \, d\Omega$$

Radiation energy

Radiation flux

$$\frac{\partial E_r}{\partial t} + \nabla \cdot \mathbf{F}_r = \text{source terms} \qquad \frac{1}{c^2}\frac{\partial \mathbf{F}_r}{\partial t} + \nabla \cdot \mathbf{P}_r = \text{source terms}$$

$$\mathbf{F}_r \propto -\nabla E_r$$ FLD!

Kuiper et al. 2011, Flock et al. 2013, Robinson et al. 2024, Ziampras et al. 2024

Moment methods: FLD and M1

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Radiation energy   Radiation flux

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$$\frac{1}{c^2}\frac{\partial \mathbf{F}_r}{\partial t} + \nabla \cdot \mathbf{P}_r = \text{source terms}$$

evolves both $E_r, \mathbf{F}_r$

closure relation for $\mathbf{P}_r$

M1!

Melon Fuksman et al. 2021, Muley et al. 2023

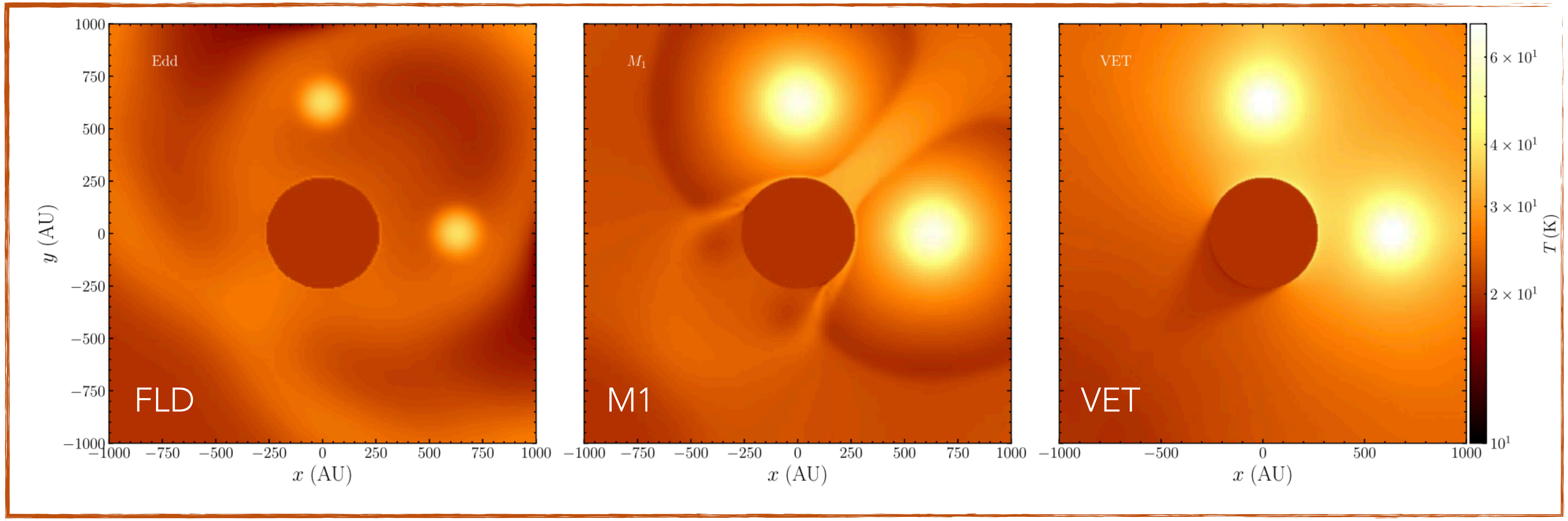
Discrete ordinates

$$\frac{1}{c}\frac{\partial I_i}{\partial t} + \mathbf{n}_i \cdot \nabla I_i = \eta - \chi I_i$$

solve for intensity in a finite set of directions $\mathbf{n}_i$

Jiang et al. 2021, Baronett et al. 2026

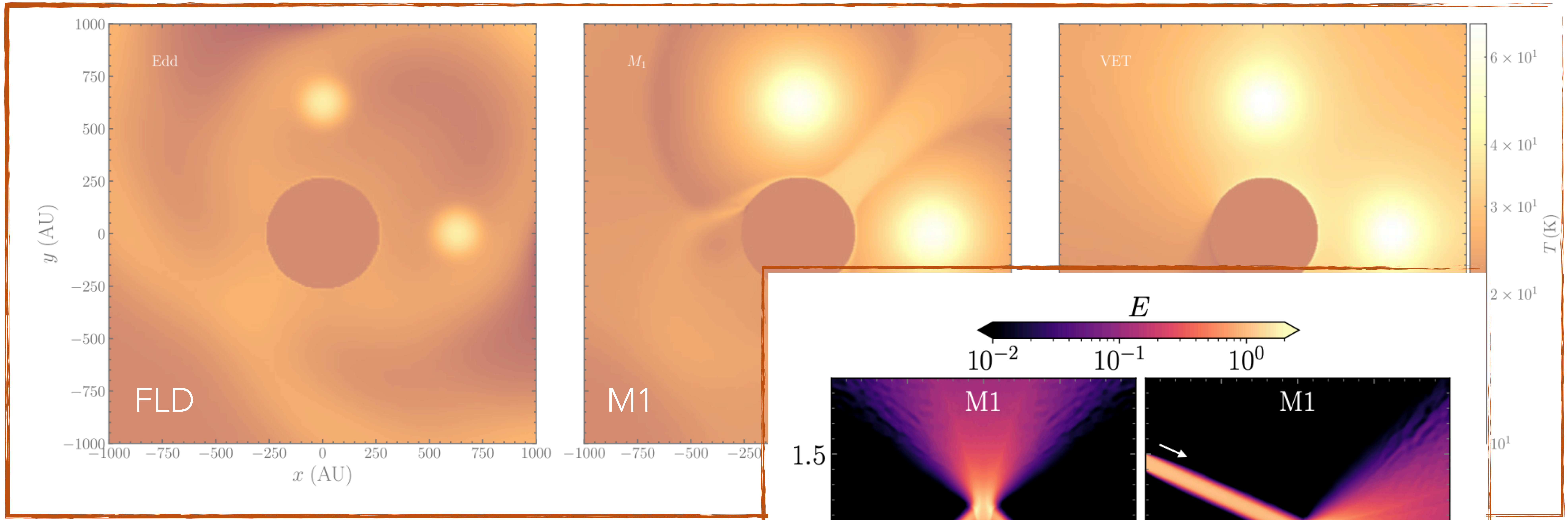
Radiation Shadow

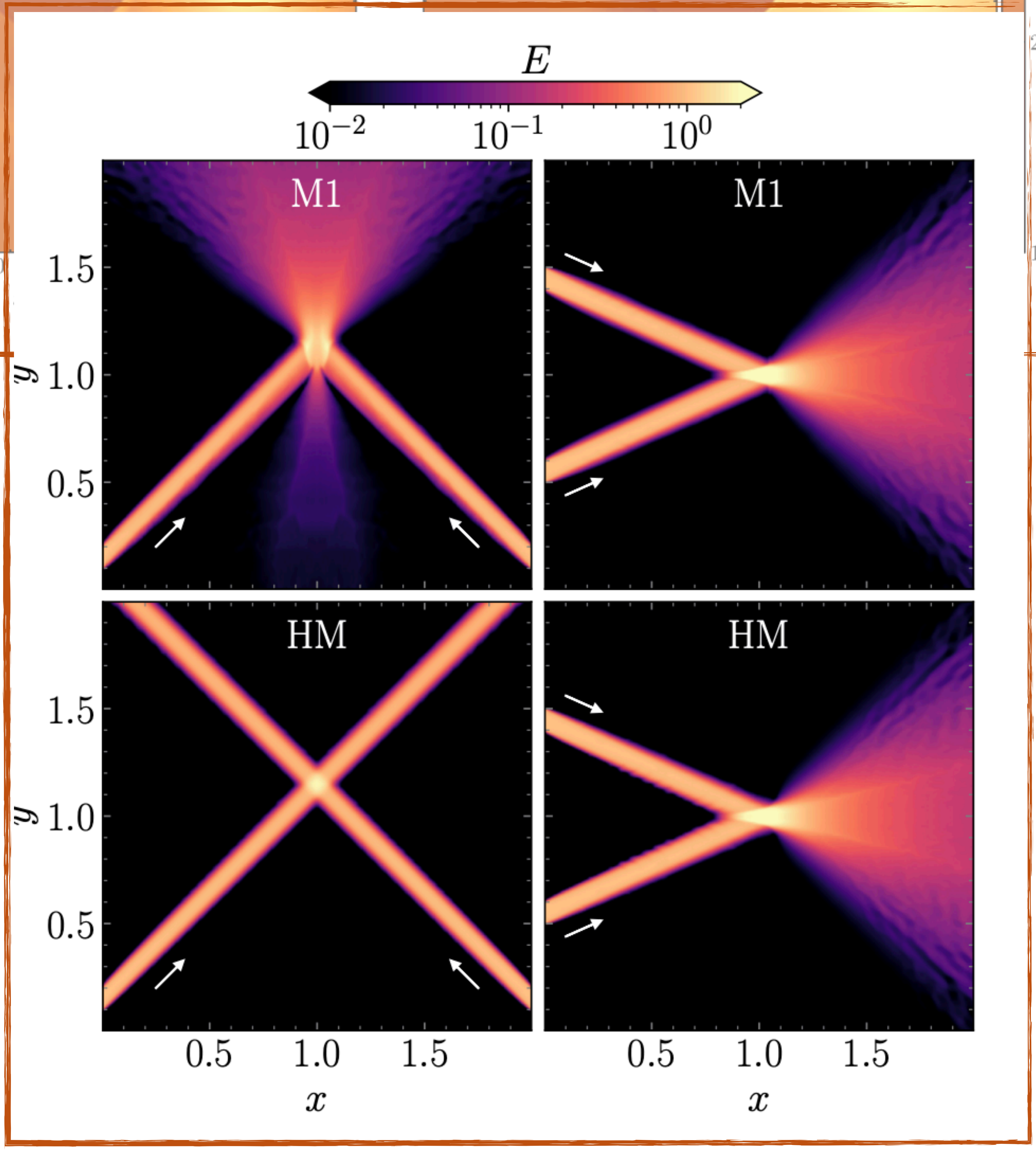


two diffuse spherical sources of radiation shining on an opaque clump in an optically thin medium

Menon et al. 2022

Crossing beams



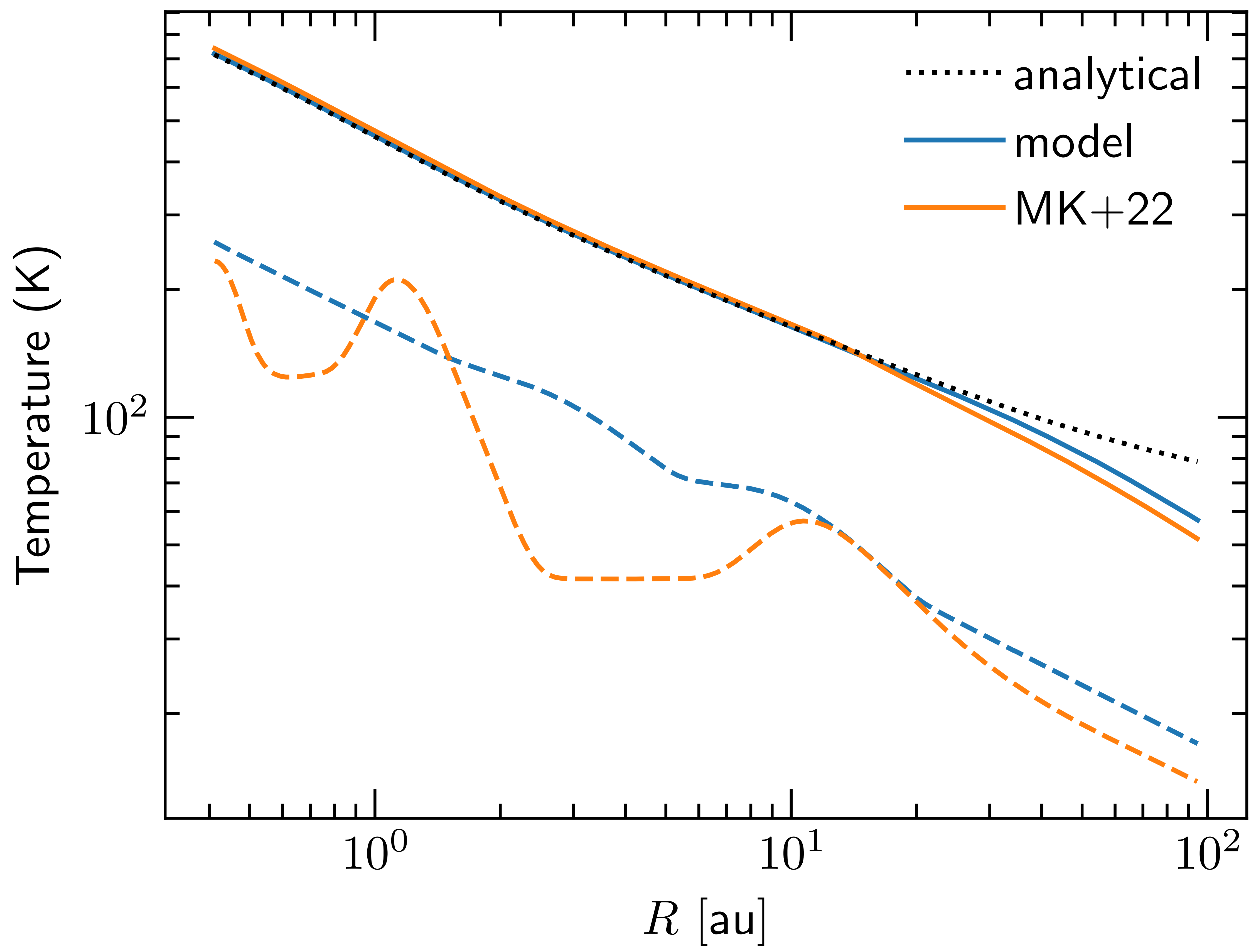


FLD: no beam crossing, they smear and diffuse out

M1: can handle a single beam, but they don't cross

Menon et al. 2022, Melon Fuksman et al. 2025

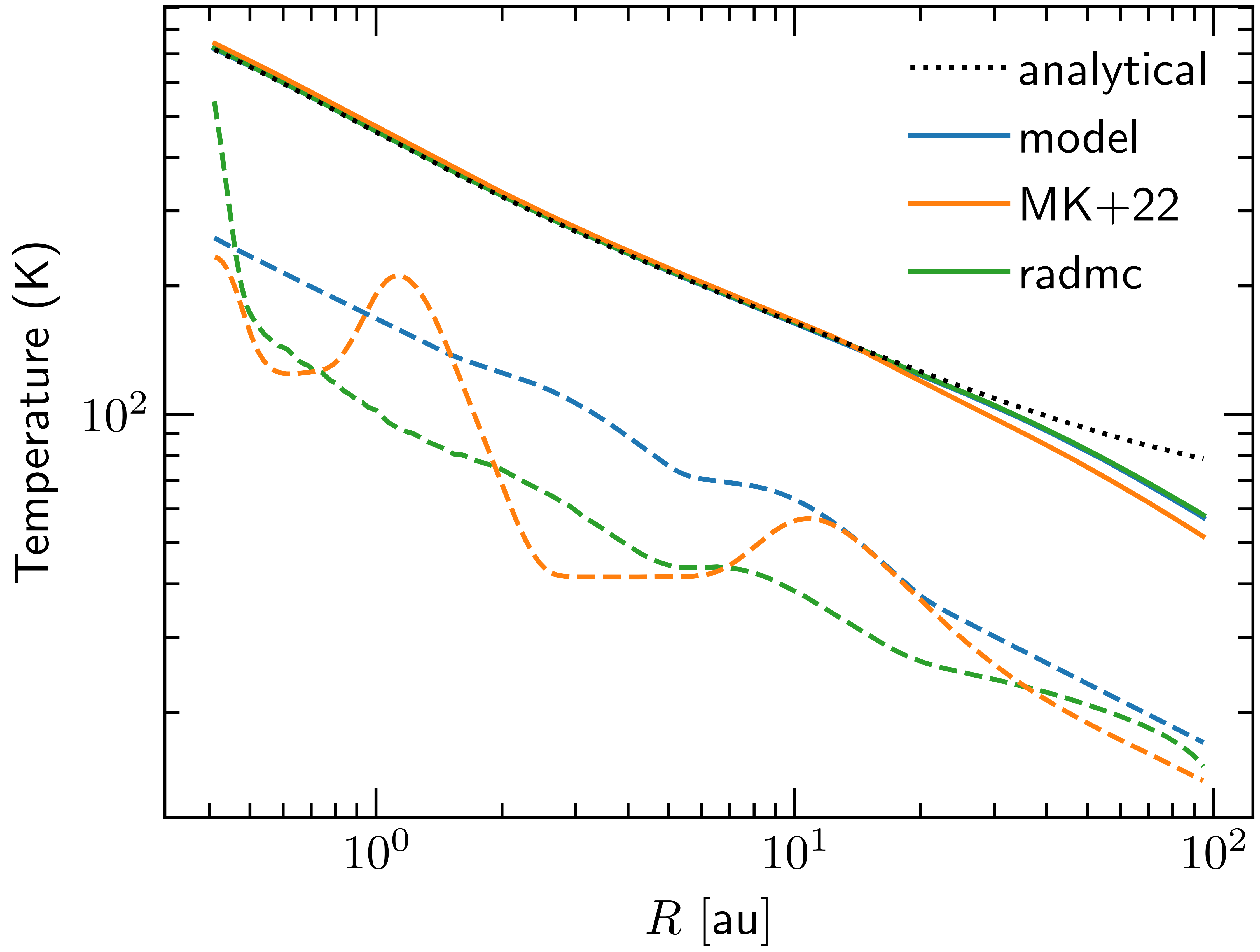
M1 vs FLD for self-shadowed disks: hydrostatics



- great agreement at the **surface** (Chiang & Goldreich 1997)
- temperature bumps: **lower amplitudes** with **FLD** but overall not treating (**vertical + in-plane**) radiation transport resulted in "exaggerated" wave amplitudes
- Both FLD and M1 show **higher midplane** temperatures compared to MCRT

Sudarshan et al. 2026 (with M1 data from Melon Fuksman & Klahr 2022)

M1 vs FLD for self-shadowed disks: hydrostatics



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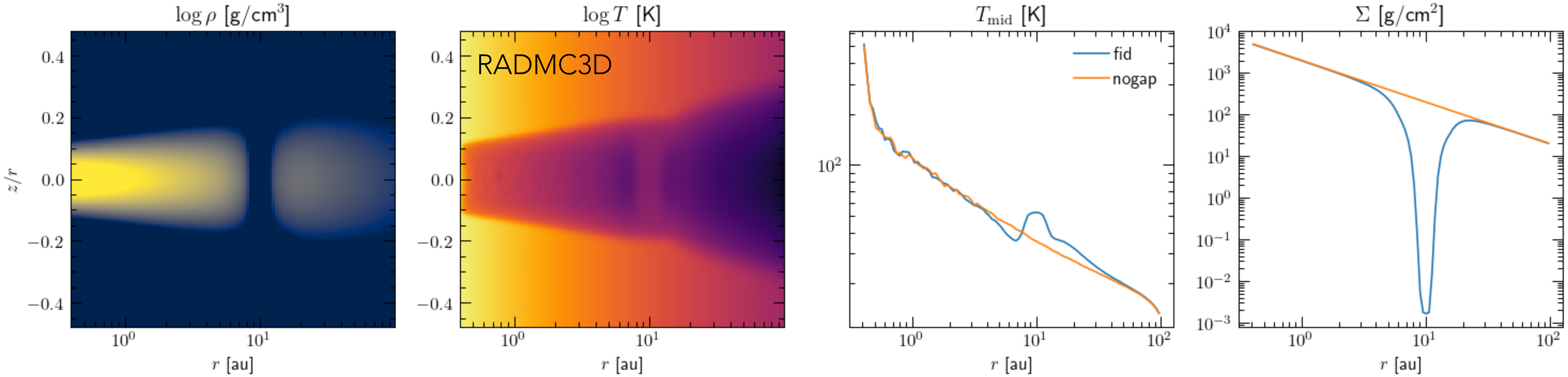
Comparison: temperatures around a planet gap

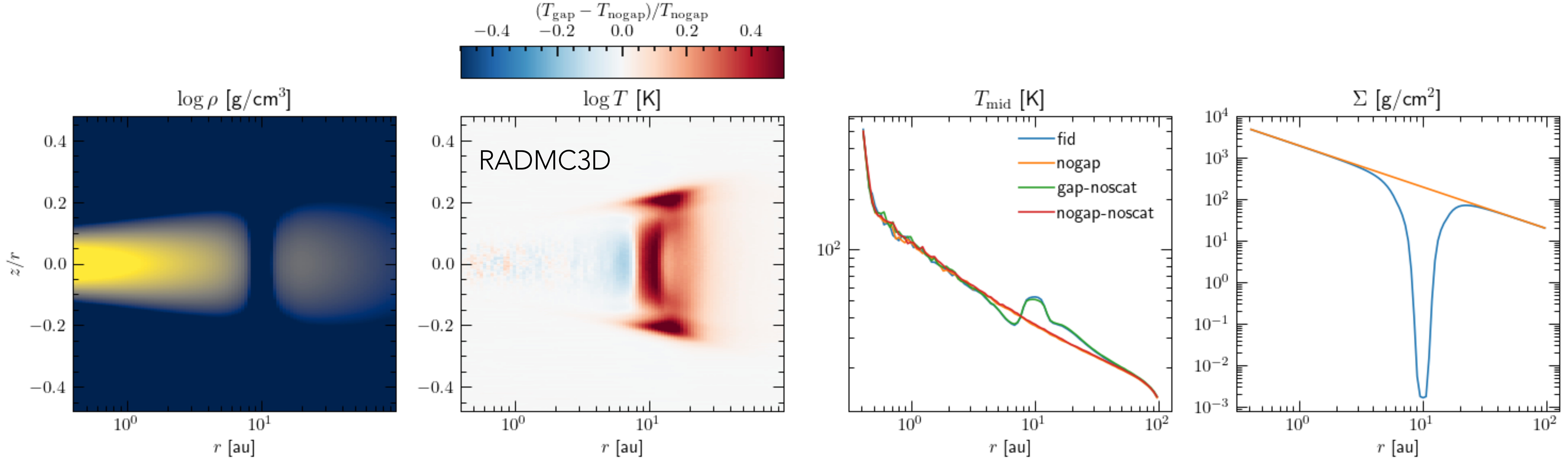
Control: RADMC setup, standard disk with a duffel gap (10 M_{jup}, at 10 au)



- The gap region in the mid plane is warmer, than the background disk.

Comparison: temperatures around a planet gap

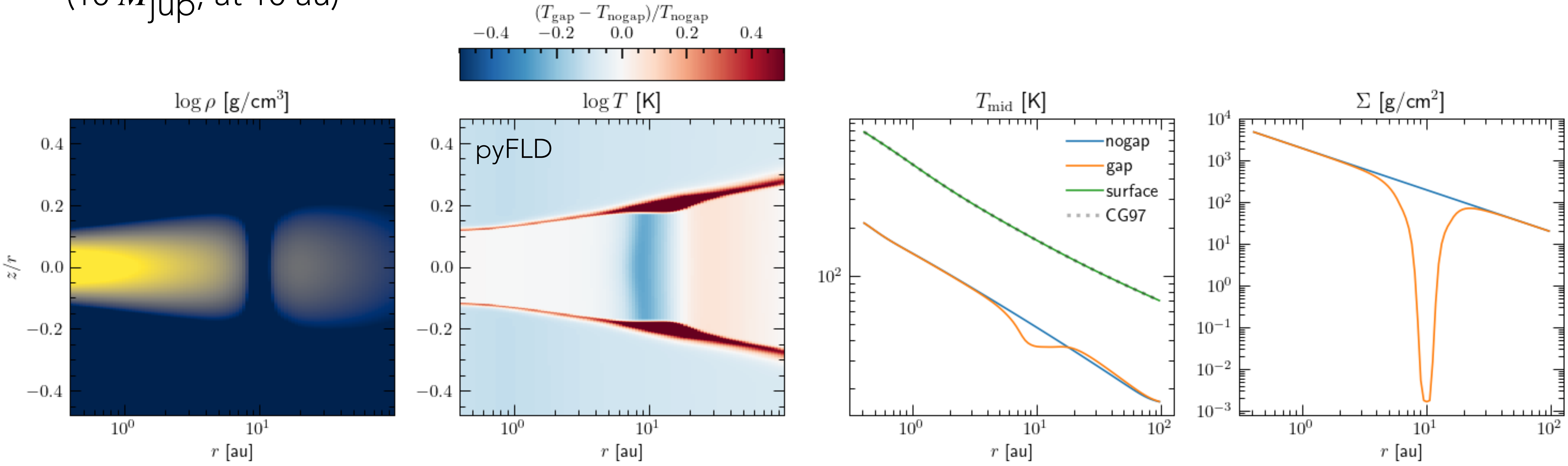
Control: RADMC setup, standard disk with a duffel gap (10 M_{jup}, at 10 au)



- By about 40%. This result is also consistent, with/without scattering in RADMC-3D (scattering_mode=2) for our opacity model.

with moment methods:

Corresponding FLD setup with pyFLD (by Alex Ziampras), standard disk with a duffel gap ($10\ M_{\rm jup}$, at 10 au)

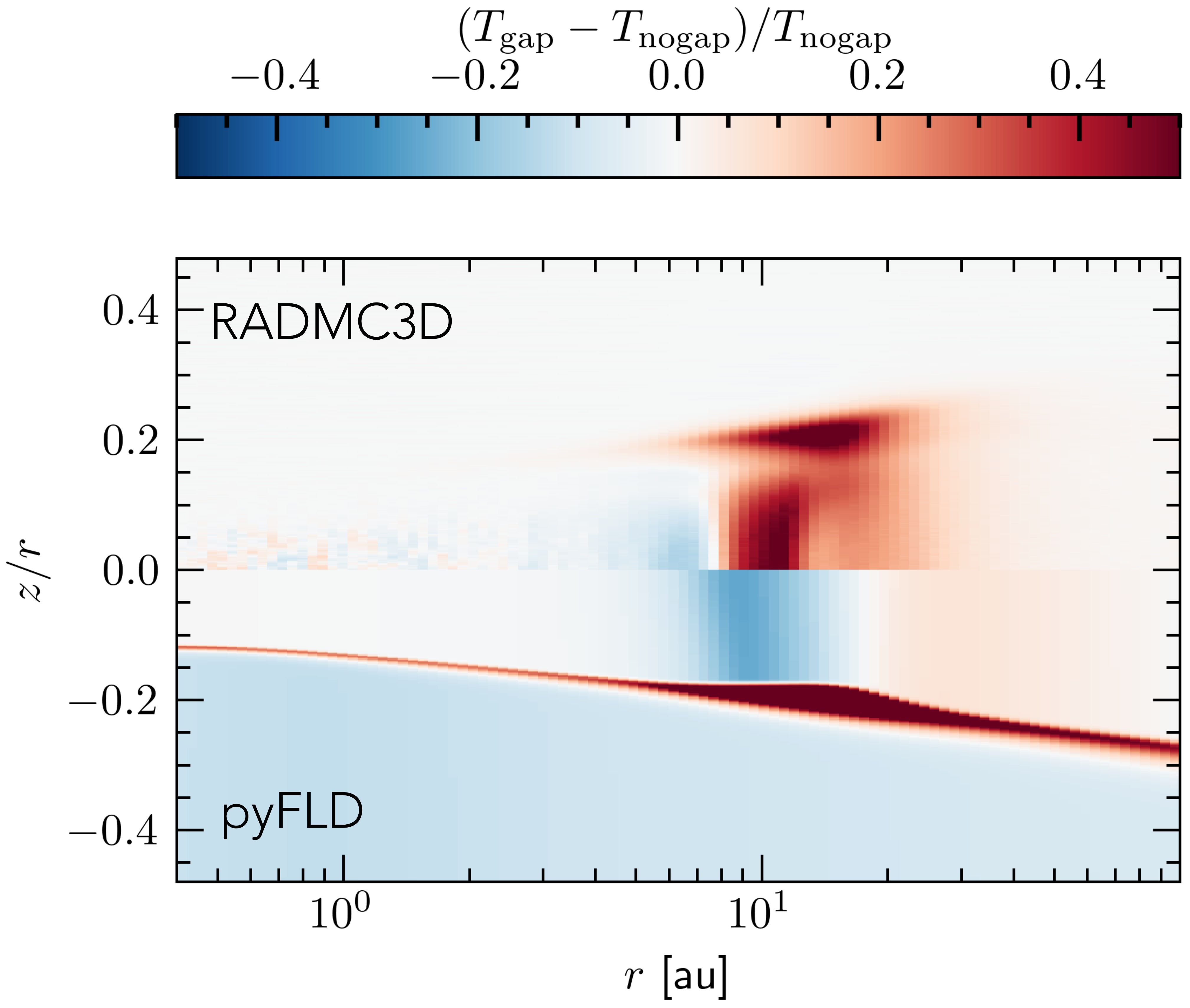


- the atmosphere matches the standard Chiang & Goldreich optically thin solution
- differences at the disk midplane, FLD (and the other moment methods) struggle to capture the warm gap

RADMC vs FLD: temperatures around a planet gap

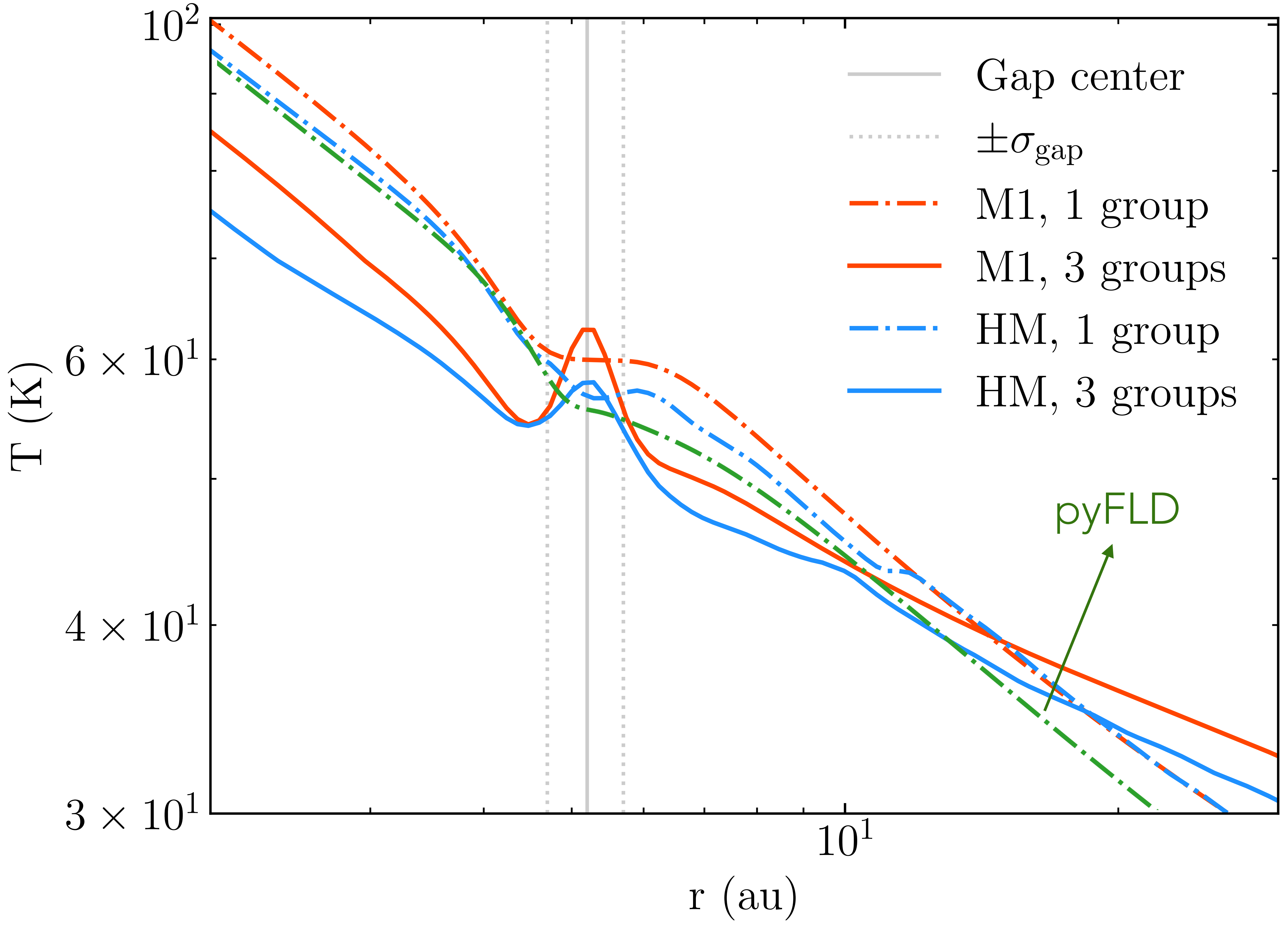
- The likely story for what FLD misses:
  - re-emission from the gap edge

Heated gap edge → emits IR over wavelengths where opacity is different → radiation leaks through and hence you get a warm gap



Comparison against M1 and half-moment methods

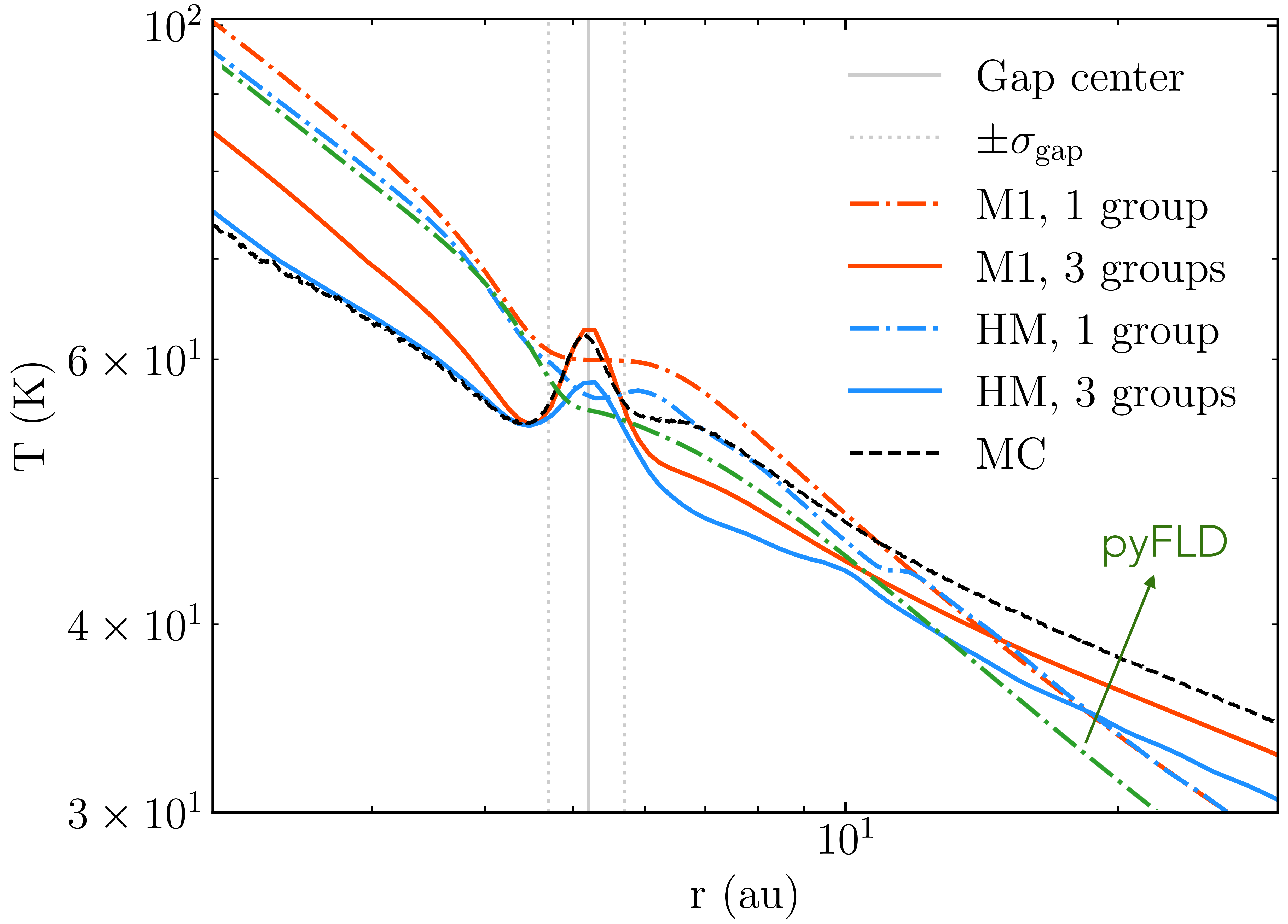
- Gray (1 frequency group) RT with moment methods seem to all show a "colder" gap
- Things start improving as you go towards multi-group methods!



M1 and HM simulations by David Melon Fuksman!

FLD, M1, HM and RADMC3D

- M1 with 3 groups seems to be the best match within the gap, but it worsens for the rest of the disk



We will also be including the discrete ordinate method
- thanks to Stanley Baronett and Zhaohuan Zhu!

Outlook

- It can be difficult to detangle what effects change with different radiation methods: comparisons are essential
- Standard tests are useful, but we can now start comparing realistic "disk" setups between the methods
- First results: moment methods don't end up capturing the warmer gap temperatures in disks, but this improves if you start including frequency dependence (multi-group) methods
- More diagnostics underway, we're also including more codes: freq- dependent FLD (cuDisc), discrete ordinates (Athena++)